Before LLM

10 BASIC NLP

techniques

Regular Expressions (Regex)

- The Swiss Army knife of text processing
- Lightning fast, processes millions of strings in seconds
- Zero model loading, zero inference cost
- Perfect for: email extraction, phone validation, log parsing
- Pro tip: Vibe Regex + strong unit tests = Works well
- When NOT to use: Anything requiring semantic understanding

Bag of Words (BoW)

- Your first baseline - never skip this step
- Converts text to word frequency vectors
- Surprisingly effective for spam detection & sentiment
- Fully interpretable - explain every prediction
- Scales to millions of documents easily
- Combine with: Logistic Regression or Naive Bayes
- Cost: Nearly zero compute, instant training

TF-IDF

- BoW's smarter cousin
- Downweights common words, highlights unique terms
- Still powers many production search systems
- Great for: keyword extraction, document similarity
- Works beautifully with cosine similarity
- Training time: Seconds on large corpora
- My go-to for quick document clustering projects

n-grams

- Captures word sequences (bigrams, trigrams)
- "New York" becomes ONE feature, not two
- Use WITH BoW/TF-IDF for instant accuracy boost
- Essential for: phrase detection, language modeling
- Trade-off: More n → More features → More sparsity
- Sweet spot: Unigrams + Bigrams for most tasks

BM25

- TF-IDF's evolved form for ranking
- The algorithm behind Elasticsearch's default scorer
- Handles document length normalization smartly
- Outperforms TF-IDF for search & retrieval tasks
- Used by: Wikipedia, countless production search engines
- Combine with: Dense vectors for Hybrid Search
- My recommendation: Start here before semantic search

Word2Vec

- Words represented as 300-dimensional vectors
- King - Man + Woman = Queen (actual math!)
- Pre-trained models available — no training needed
- Captures semantic relationships efficiently
- Use for: word similarity, feature engineering
- Limitation: One vector per word (no context)
- Still useful: Initializing embedding layers

FastText

- Word2Vec + subword magic (advanced word2vec)
- Handles typos and rare words gracefully
- Perfect for: German, Turkish, morphologically rich languages
- Can generate vectors for unseen words
- Built-in text classifier and surprisingly good
- Training: Minutes on commodity hardware
- Meta's gift to practical NLP

Latent Semantic Analysis (LSA)

- SVD applied to term-document matrices
- Finds hidden semantic relationships
- Reduces dimensionality while preserving meaning
- Use for: Document clustering, semantic similarity
- Handles synonyms naturally
- Fast and deterministic
- Predecessor to modern topic models

Latent Dirichlet Allocation (LDA)

- The OG topic modeling algorithm
- Discovers themes across document collections
- Fully unsupervised, no labels needed
- Interpretable topics for stakeholder presentations
- Use for: Content categorization, trend analysis
- Pro tip: Use coherence scores to pick topic count
- Still my choice for explainable topic discovery

Vader Sentiment

- Rule-based sentiment - zero training required
- Built specifically for social media text
- Understands emojis 😃👎, slang, and punctuation!!!
- Handles CAPS for emphasis (AMAZING vs amazing)
- Use for: Tweet analysis, review monitoring, quick sentiment baseline
- Bonus: Handles negations ("not good" = negative)
- Limitation: English only, no context understanding
- Pro tip: Use as first filter before expensive LLM calls
- My take: Still unbeatable for social media sentiment at scale

Before reaching for GPT, ask:

- Can Regex solve this?
- Will TF-IDF + Logistic Regression work?
- Is BM25 enough for my search?

Simple solutions = Lower cost, faster inference, easier debugging

8+ years lesson: The best model is the one that ships.